

WebLearn Academy

Visual Project Documentation

Interactive Web Learning Platform

Built with Node.js, Express, and Vanilla JavaScript

Version	1.0.0
Technology	Node.js + Express + Vanilla JS
Database	JSON File Storage
Auth	JWT (JSON Web Tokens)
UI Theme	Glassmorphism

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1. System Architecture Diagram

The following diagram shows how the different components of WebLearn Academy interact with each other:

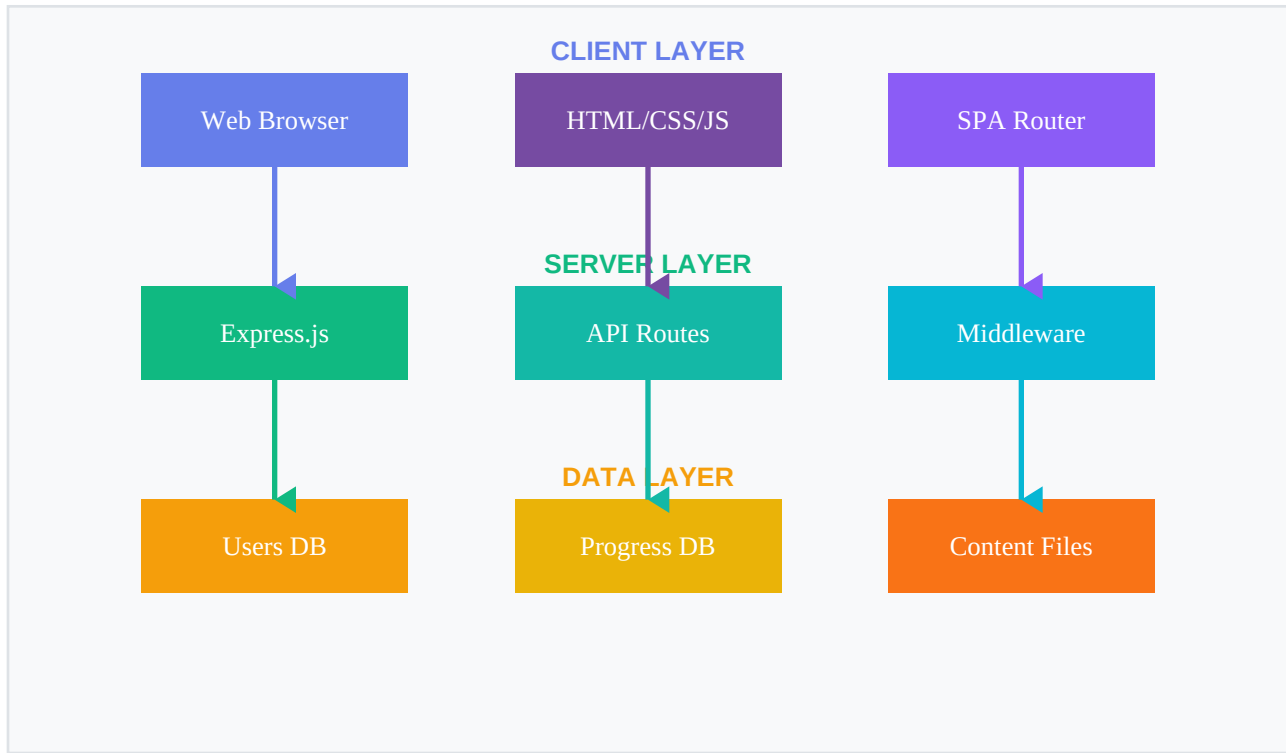


Figure 1: System Architecture - Three-tier architecture with Client, Server, and Data layers

The application follows a classic three-tier architecture. The client layer handles the user interface, the server layer processes requests and business logic, and the data layer stores all persistent information in JSON files.

2. Project Structure Visualization

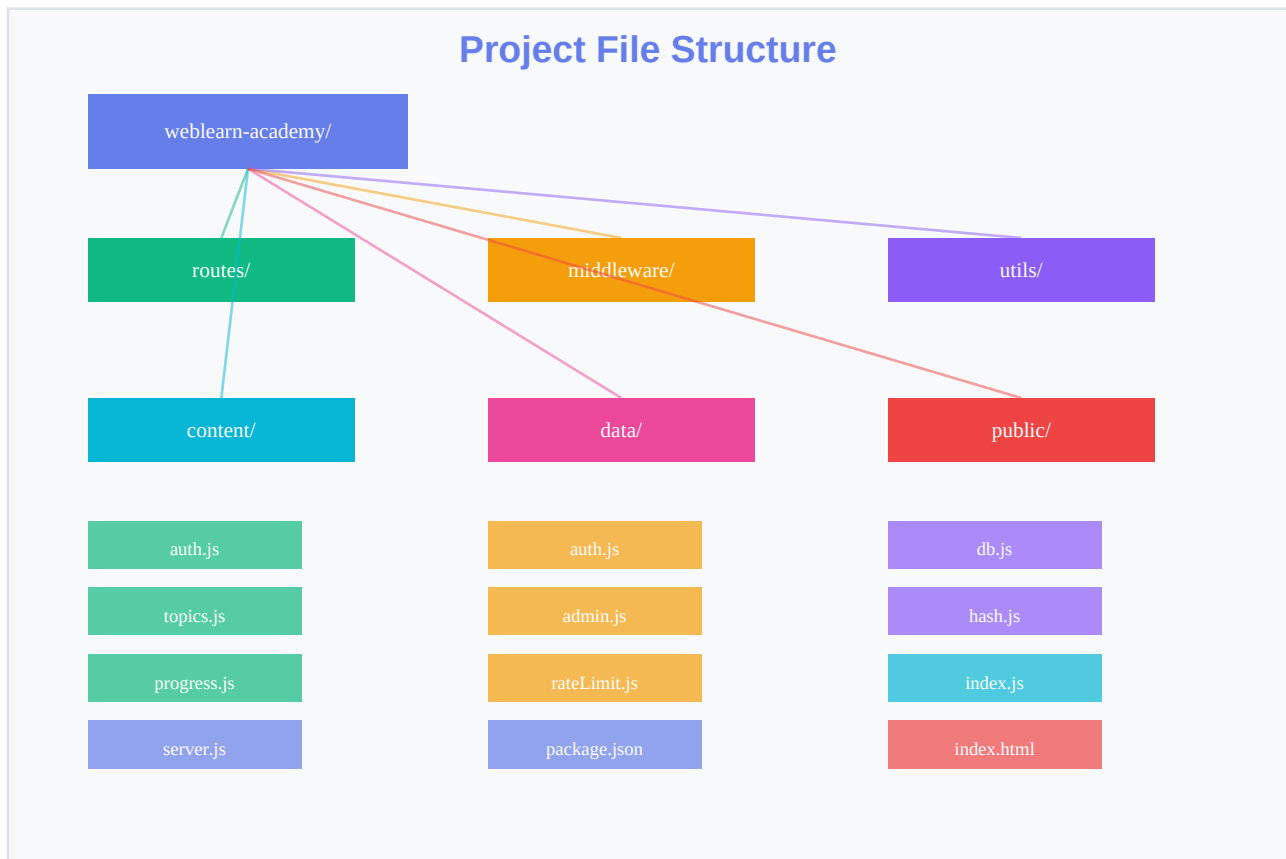


Figure 2: Project folder structure with file organization

3. Authentication Flow Diagram

The following diagram illustrates the complete authentication process from user registration to accessing protected routes:

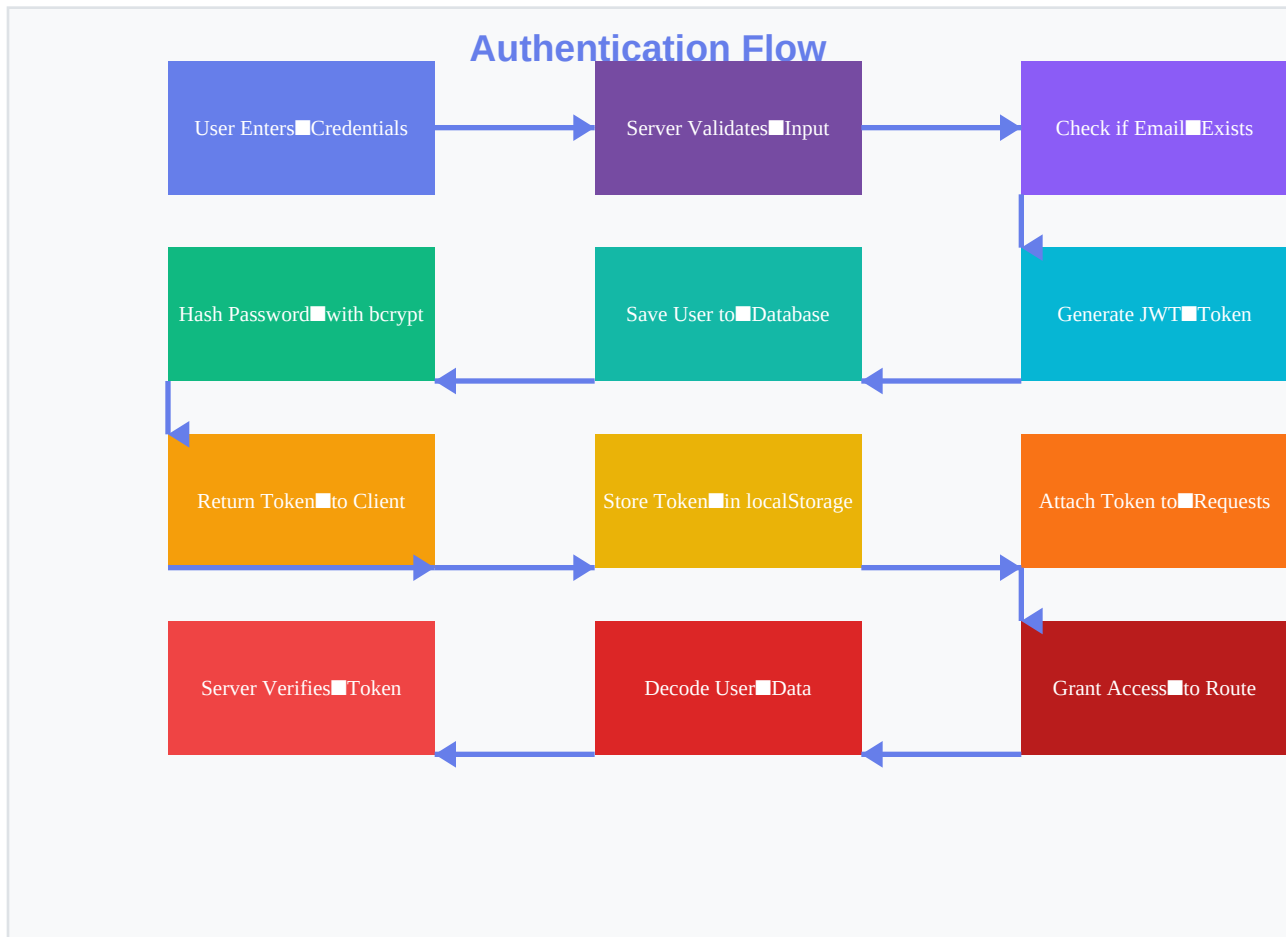


Figure 3: Complete authentication flow from login to route access

Key Security Features:

- Passwords are never stored in plain text (bcrypt hashing)
- JWT tokens expire after 7 days
- Rate limiting prevents brute force attacks
- Admin routes require additional role verification

4. Database Schema Diagram

WebLearn Academy uses JSON file-based storage. Here's the data structure:

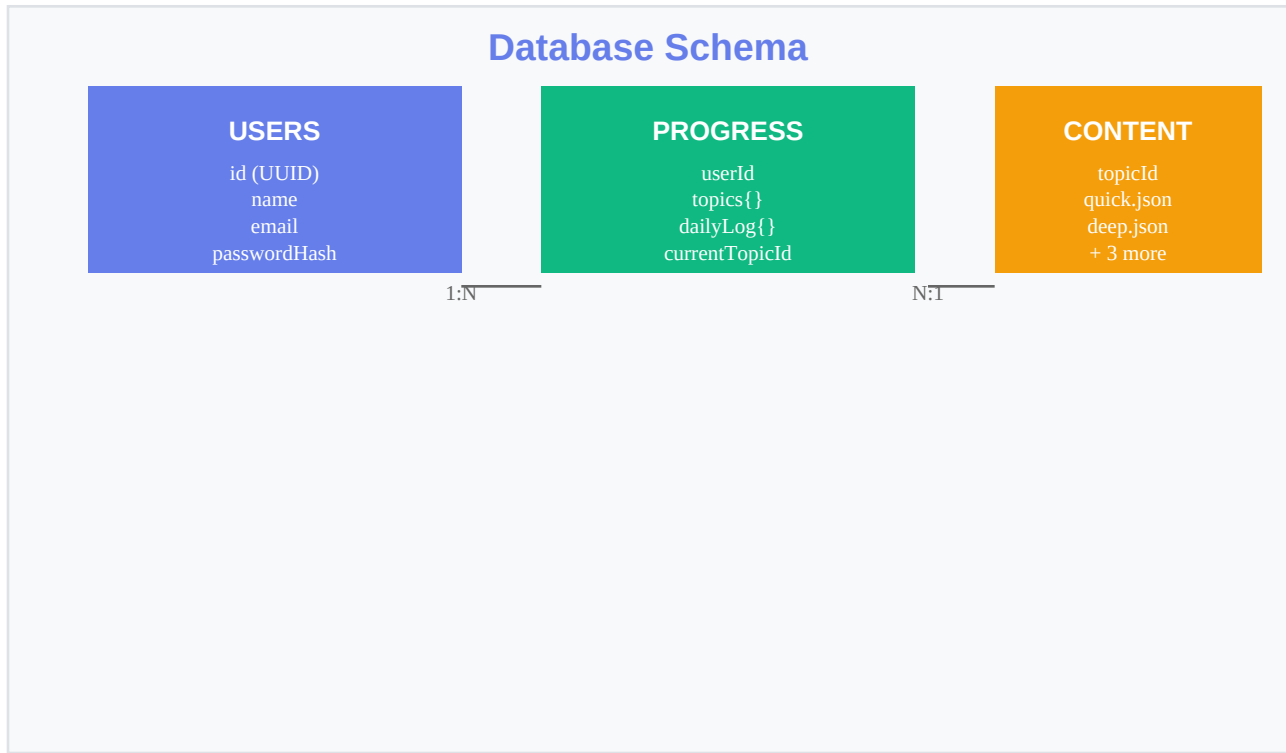


Figure 4: Database schema showing users, progress, and content relationships

Data Storage Details:

- **users.json**: Stores all user accounts with hashed passwords
- **progress.json**: Tracks learning progress for each user
- **content/**: Contains JSON files with lesson content organized by topic

5. User Interface Mockups

The application features a modern glassmorphism design. Here are the main screens:

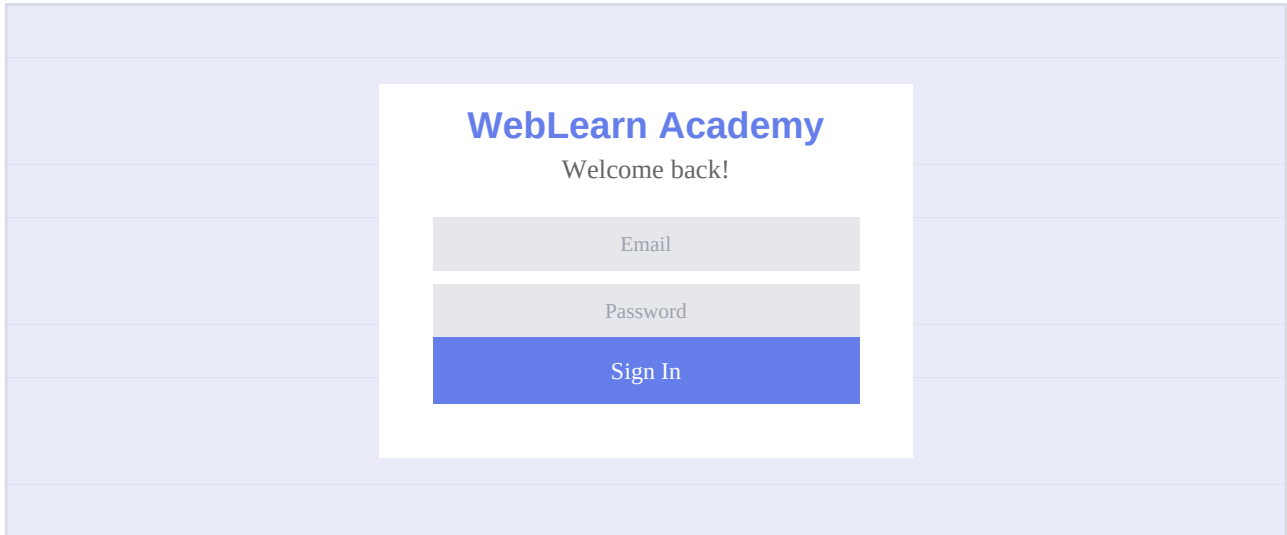


Figure 5: Login screen with glassmorphism card design

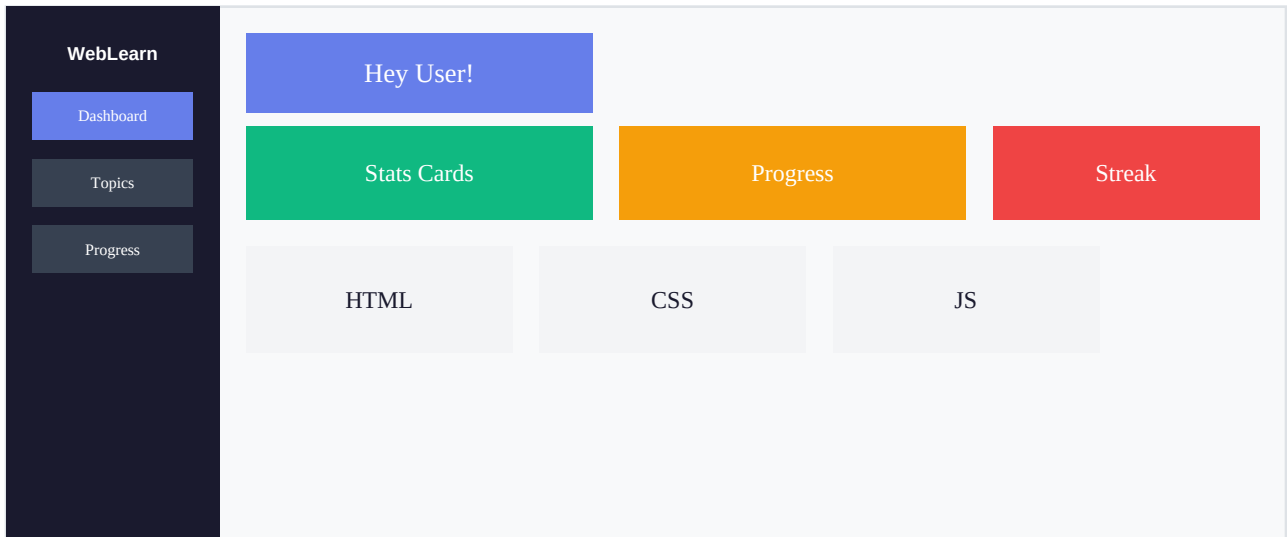


Figure 6: Customer dashboard with sidebar navigation and stats cards

6. API Endpoint Map

Visual representation of all API endpoints organized by module:

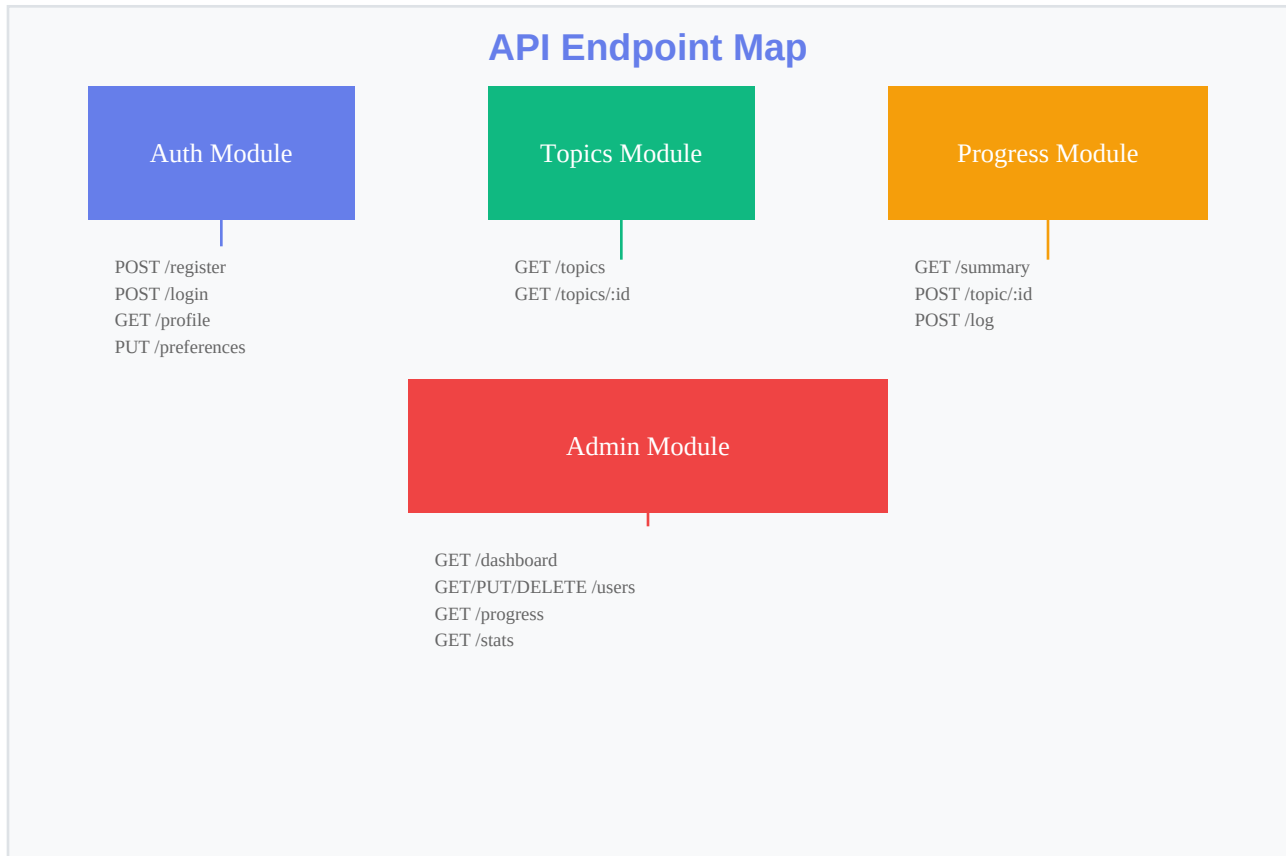


Figure 7: API endpoints organized by functional module

7. Content Structure Diagram

Each learning topic contains 5 types of content sections:

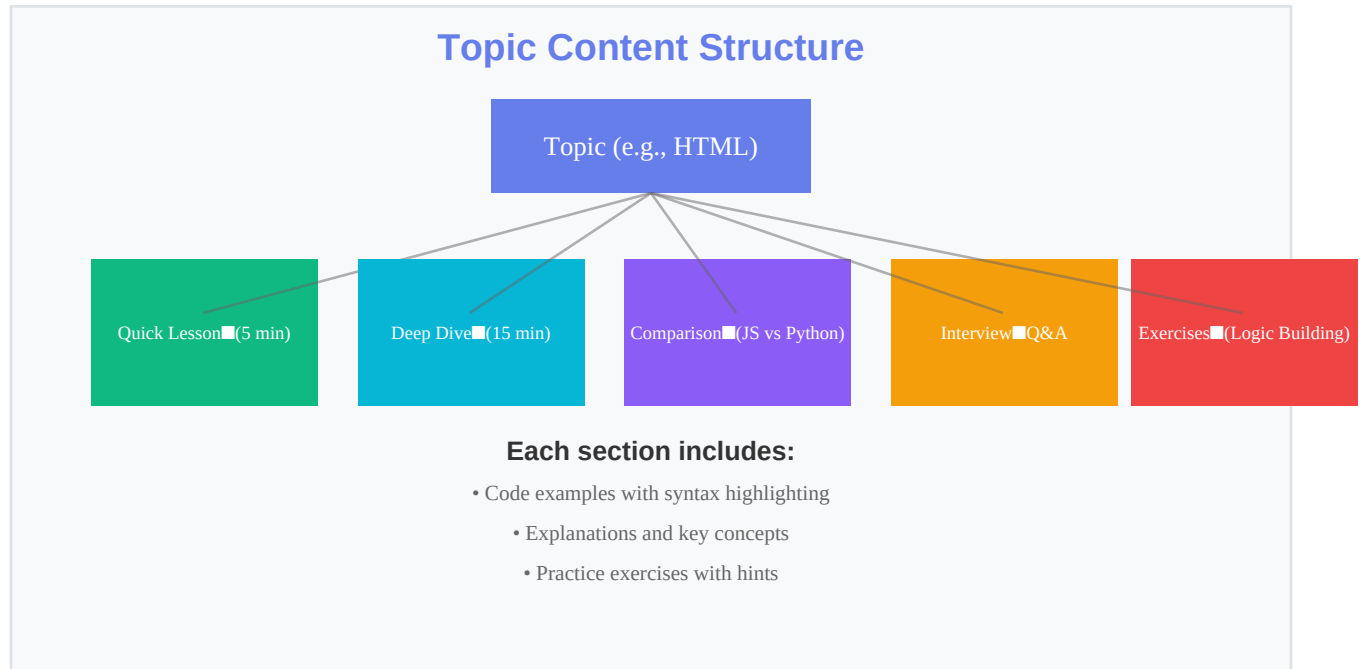


Figure 8: Topic content structure with 5 learning sections

8. Admin Panel Flow

The admin panel provides comprehensive user management capabilities:

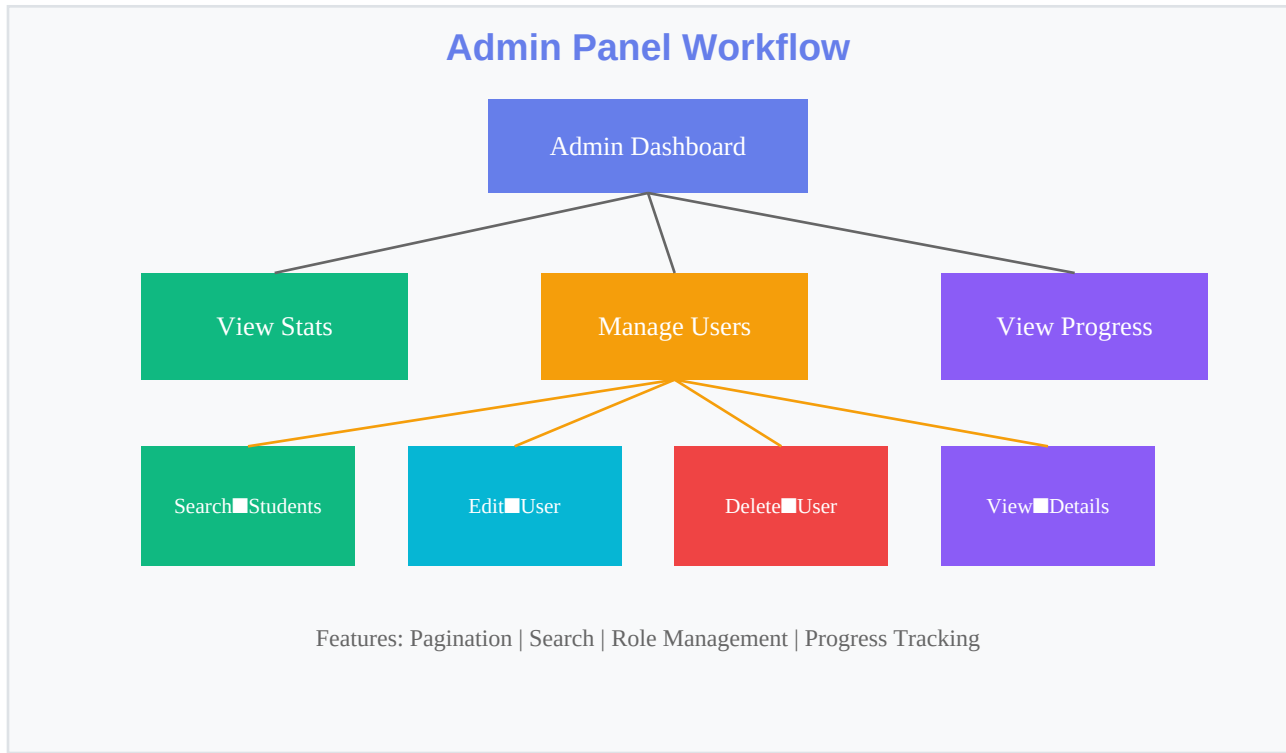


Figure 9: Admin panel workflow with user management actions

9. Learning Path Diagram

The 15-day fast-track learning path through web development topics:

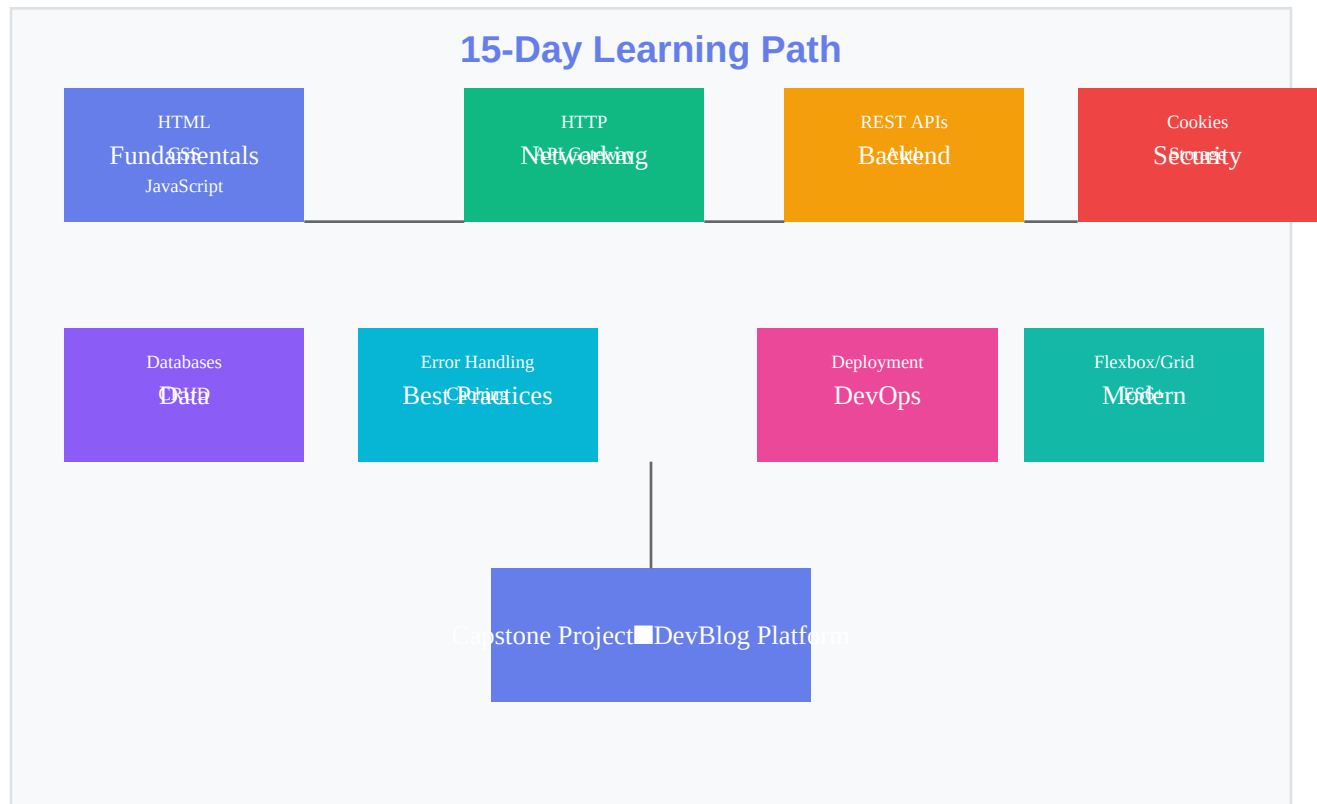


Figure 10: 15-day learning path with topic progression

10. Technology Integration

How different technologies work together in the stack:

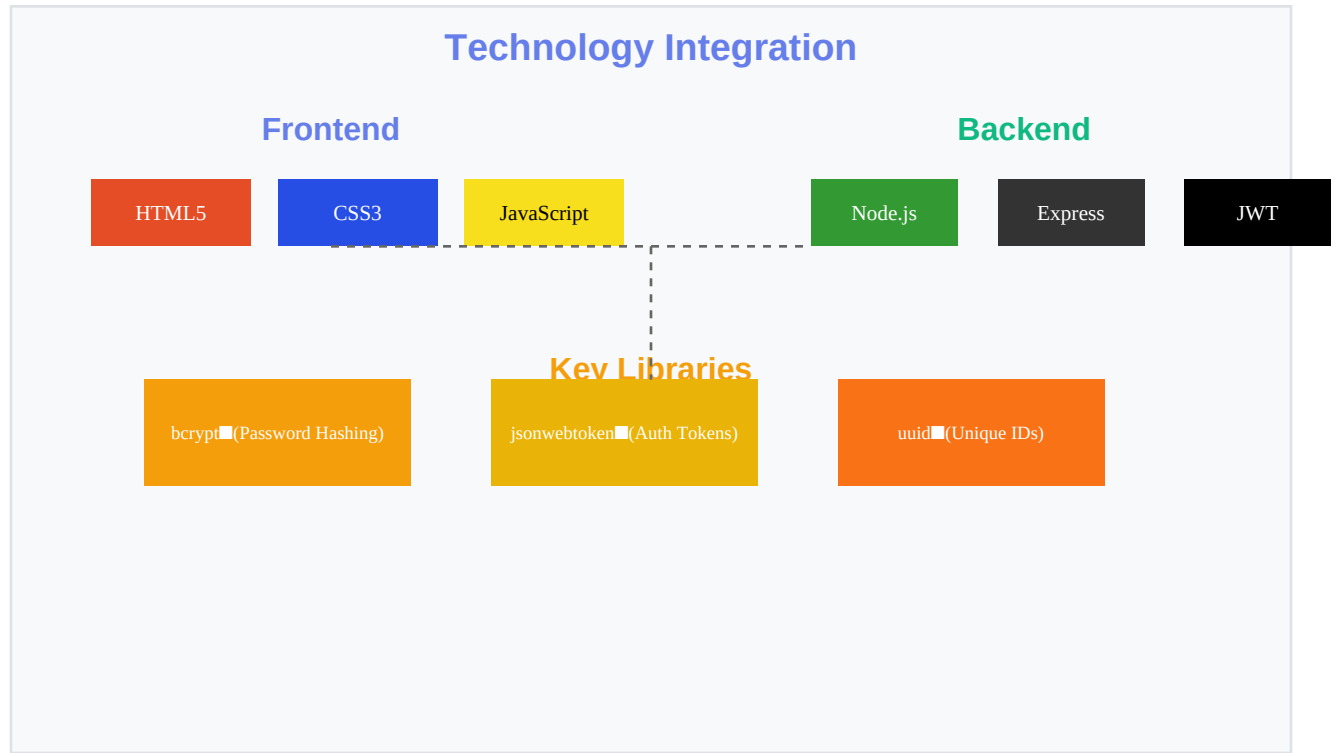


Figure 11: Technology stack integration diagram

WebLearn Academy

A complete, modern web learning platform built with simplicity and scalability in mind